Mathematical and Computational Methods

Exercise Sheet 8: Differentiation II: applications

Tutorial route (▶, in this order, about 80 minutes): 1.1, 2.4, 2.7, 2.6(a–d), 2.8, 2.10(a,c), 2.2(a,c). Everything else is for home study.

Levels: ⋆ drill, ⋆⋆ core, ⋆⋆⋆ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Exercise 1.1 *(▶ ⋆ drill, tangents & normals)*. For $f(x) = x^2$, find the equations of the tangent and the normal to the curve at the point $(2, 4)$.

Exercise 1.2 *(⋆ drill, intermediate value theorem)*. Let $f(x) = x^3 + x - 1$.

(a) Show that $f(x) = 0$ has a solution in the interval $(0, 1)$.

(b) Show that this is the *only* real solution. (Hint: the sign of f'.)

Exercise 1.3 *(⋆ drill, l'Hôpital)*. Evaluate, stating that the form is indeterminate before applying the rule:

$$\text{(a) } \lim_{x\to 0}\frac{\sin x}{x}, \qquad \text{(b) } \lim_{x\to 0}\frac{e^x - 1}{x}, \qquad \text{(c) } \lim_{x\to\infty}\frac{\ln x}{x^2}.$$

Exercise 1.4 *(⋆ drill, classify)*. Locate and classify the stationary points of $f(x) = x^3 - 3x^2$ using the second-derivative test.

Exercise 1.5 *(⋆ drill, asymptotes)*. Find the vertical and horizontal asymptotes of

$$\text{(a) } f(x) = \frac{1}{x-2}, \qquad \text{(b) } g(x) = \frac{2x}{x-3}.$$

Exercise 1.6 *(⋆ drill, extreme-value / closed interval)*. Use the closed-interval (extreme-value-theorem) method to find the absolute maximum and minimum of $f(x) = x^2 - 4x + 1$ on $[0, 3]$.

Exercise 1.7 *(⋆ drill, optimisation — spot the error)*. Two non-negative numbers add to 10, and we want their largest possible product. A student writes the numbers as x and $10 - x$, sets $P(x) = x(10 - x)$, solves $P'(x) = 10 - 2x = 0$ to get $x = 5$, and concludes "$P''(5) = -2 < 0$, so this is a minimum." Identify the error, then find the largest product, checking the endpoints.

Exercise 1.8 *(⋆ drill, gradient descent)*. Apply gradient descent to $f(x) = x^2$ with learning rate $\eta = 0.1$, starting from $x_0 = 1$. Compute x_1 and x_2, and state the limit of the iteration.

2 Tutorial problems

Exercise 2.1 *(⋆⋆ core, tangents & normals)*. Consider the curve $y = x^3 - 2x + 1$.

(a) Find the tangent and normal lines at the point where $x = 1$.

(b) Find the equations of *all* tangent lines to the curve that are parallel to the line $y = x$.

Exercise 2.2 *(▶ ⋆⋆ core, l'Hôpital — several forms)*. Evaluate, first rewriting each into a $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form where necessary:

$$\text{(a)}\ \lim_{x\to 0}\frac{x-\sin x}{x^3}, \qquad \text{(b)}\ \lim_{x\to 0^+} x\ln x, \qquad \text{(c)}\ \lim_{x\to 0}\left(\frac{1}{x}-\frac{1}{\sin x}\right).$$

Exercise 2.3 *(⋆⋆ core, Newton's method)*. The equation $x^3 - 2x - 5 = 0$ has a single real root, which the intermediate value theorem traps in $(2, 3)$ (since $f(2) = -1 < 0$ and $f(3) = 16 > 0$). Use *Newton's method* to pin it down: starting from $x_0 = 2$, write the iteration explicitly and carry out two steps by hand, tabulating x_n, $f(x_n)$ and $f'(x_n)$ at each stage to obtain x_1 and x_2. Comment on how fast the estimate is settling.

Exercise 2.4 *(▶ ⋆⋆ core, classify — worked-example anchor)*. Locate and classify all stationary points of

$$f(x) = 2x^3 - 3x^2 - 12x + 5.$$

Exercise 2.5 *(⋆⋆ core, full polynomial sketch)*. Carry out a full curve-sketching analysis of $f(x) = x^3 - 6x^2 + 9x$ (domain/symmetry, intercepts, increase/decrease and extrema, concavity and inflection) and sketch the curve.

Exercise 2.6 *(▶ ⋆⋆ core, full rational sketch)*. Let $f(x) = \dfrac{x}{x^2-1}$.

(a) State the domain and symmetry of f, its intercepts, and its vertical and horizontal asymptotes.

(b) Find $f'(x)$ and the intervals on which f increases or decreases. Does f have any stationary points?

(c) You may quote $f''(x) = \dfrac{2x(x^2+3)}{(x^2-1)^3}$. Find where f is concave up and concave down, and any point of inflection.

(d) Sketch the curve.

Exercise 2.7 *(▶ ⋆⋆ core, optimisation — worked-example anchor)*. An open-topped box is made from a square sheet of card of side 12 cm by cutting a small square of side x from each corner and folding up the flaps. Find the value of x that maximises the volume, and the maximum volume.

Exercise 2.8 *(▶ ⋆⋆ core, physics: maximum power transfer)*. A battery of EMF $\varepsilon \neq 0$ and internal resistance $r > 0$ delivers power $P(R) = \dfrac{\varepsilon^2 R}{(R+r)^2}$ to an external load of resistance $R > 0$. Show that the power delivered is greatest when $R = r$, and find that maximum power.

Exercise 2.9 *(⋆⋆ core, data science: least squares and the mean)*. Given data values $x_1, \ldots, x_n$, the *sum of squared errors* of a single predicted constant c is $L(c) = \sum_{i=1}^{n}(x_i - c)^2$.

(a) Show that L is minimised when c equals the mean $\bar{x} = \frac{1}{n}\sum_i x_i$.

(b) Find the minimising c for the data $\{2, 4, 9\}$.

Exercise 2.10 *(▶ ⋆⋆ core, gradient descent on a quadratic)*. Minimise $f(x) = x^2 - 4x + 5$ by gradient descent with learning rate $\eta = 0.25$, starting from $x_0 = 0$.

(a) Write the update rule $x_{n+1} = g(x_n)$ explicitly and compute x_1, x_2, x_3.

(b) To what value do the iterates converge, and why is that the right answer?

(c) What would go wrong if the learning rate were taken as $\eta = 1.5$?

3 Further practice (home)

Exercise 3.1 *(⋆⋆ core, mean value theorem).*

(a) Find the value of c guaranteed by the mean value theorem for $f(x) = x^3$ on $[0, 2]$.

(b) Use the mean value theorem to prove that $|\sin a - \sin b| \le |a - b|$ for all real a, b.

Exercise 3.2 *(⋆⋆ core, inverse-function theorem).* Let $f(x) = 2x + \cos x$.

(a) Show that f is strictly increasing on $\mathbb{R}$, hence one-to-one and invertible.

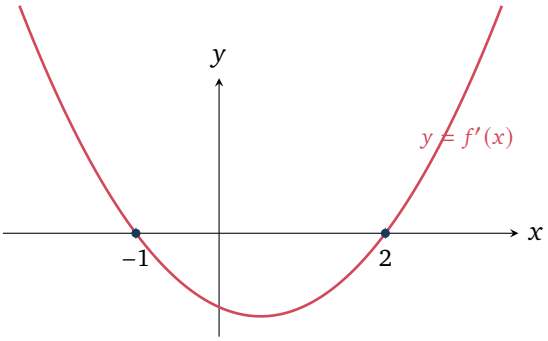
(b) Compute $\left(f^{-1}\right)'(1)$.

Exercise 3.3 *(⋆⋆ core, classify a mixed bag).* Locate and classify all critical points of:

(a) $f(x) = x^3 - 3x^2 + 3x$, (b) $g(x) = x + \dfrac{1}{x}$, (c) $h(x) = xe^{-x}$, (d) $k(x) = x^{2/3}$.

In (d), show first that $k'(0)$ does not exist.

Exercise 3.4 *(⋆⋆ core, interpret a graph of f').* The graph below shows the *derivative* f' of a function f (not f itself). Assume that outside the window f' continues as the picture suggests: it has no further zeros, it keeps falling to the left and it keeps rising to the right. Using only this information, describe f: state the intervals on which f is increasing or decreasing, locate and classify its stationary points, and say where f is concave up or down and where it has a point of inflection.



Exercise 3.5 *(⋆⋆ core, physics: kinematics min/max).* A particle moves along a straight line with position $s(t) = t^3 - 9t^2 + 24t$ metres at time t seconds, for $0 \le t \le 5$. Find (a) the times at which it is instantaneously at rest, and (b) its maximum and minimum velocity on the interval.

Exercise 3.6 *(⋆ drill, true or false).* For each statement, say whether it is true or false and justify briefly.

(a) If $f'(c) = 0$ then f has a local maximum or minimum at c.

(b) If $f''(c) = 0$ then c is an inflection point of f.

(c) A continuous function on an open interval need not attain a maximum value.

(d) If $f'(x) > 0$ on an interval, then f is strictly increasing there.

(e) At a point where $f'(a) = 0$, the normal line is horizontal.

(f) l'Hôpital's rule can be applied to any limit of a quotient.

Exercise 3.7 *(⋆ ⋆ ⋆ challenge, challenge: a chord property).* Show that the tangent to $y = x^3$ at the point (a, a^3) (with $a \neq 0$) meets the curve again at exactly one further point, and that its x-coordinate is $-2a$.

Exercise 3.8 *(⋆ ⋆ ⋆ challenge, challenge: the Norman window)*. A "Norman" window is a rectangle topped by a semicircle, the semicircle's diameter being the top of the rectangle. For a fixed perimeter P, find the proportions that maximise the area, and show that at the optimum the rectangle's height equals the radius of the semicircle.